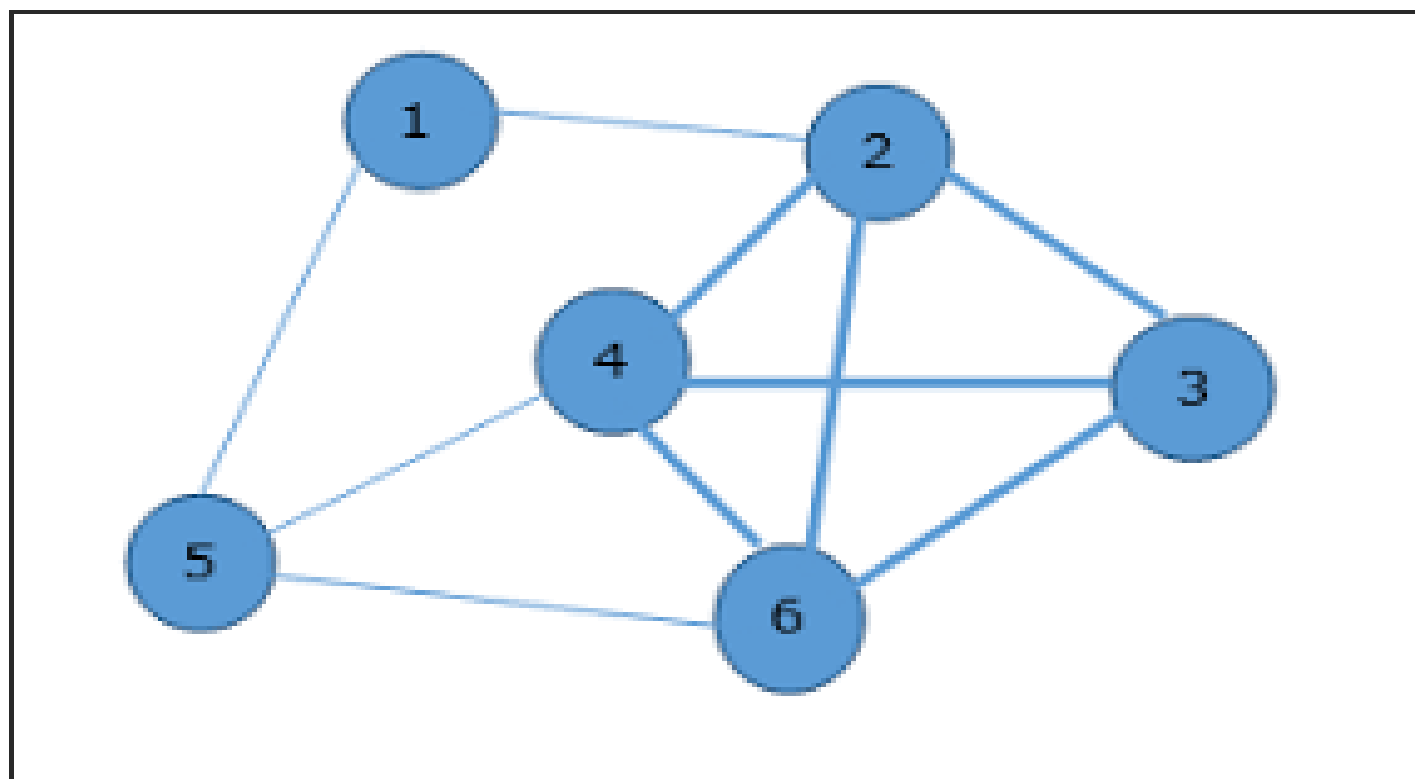


The clique problem

The clique problem

A *clique* in an undirected graph $G = (V, E)$ is a subset $V' \subseteq V$ of vertices, each pair of which is connected by an edge in E . In other words, a clique is a complete subgraph of G . The *size* of a clique is the number of vertices it contains. The *clique problem* is the optimization problem of finding a clique of maximum size in a graph. As a decision problem, we ask simply whether a clique of a given size k exists in the graph. The formal definition is

$\text{CLIQUE} = \{ \langle G, k \rangle : G \text{ is a graph containing a clique of size } k \} .$



The clique problem is NP-complete.

Proof To show that CLIQUE $\in$ NP, for a given graph $G = (V, E)$, we use the set $V' \subseteq V$ of vertices in the clique as a certificate for G . We can check whether V' is a clique in polynomial time by checking whether, for each pair $u, v \in V'$, the edge (u, v) belongs to E .

We next prove that 3-CNF-SAT $\leq_P$ CLIQUE, which shows that the clique problem is **NP-hard**.

The reduction algorithm begins with an instance of 3-CNF-SAT. Let $\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_k$ be a boolean formula in 3-CNF with k clauses. For $r = 1, 2, \dots, k$, each clause C_r has exactly three distinct literals l_1^r, l_2^r , and l_3^r . We shall construct a graph G such that ϕ is satisfiable if and only if G has a clique of size k .

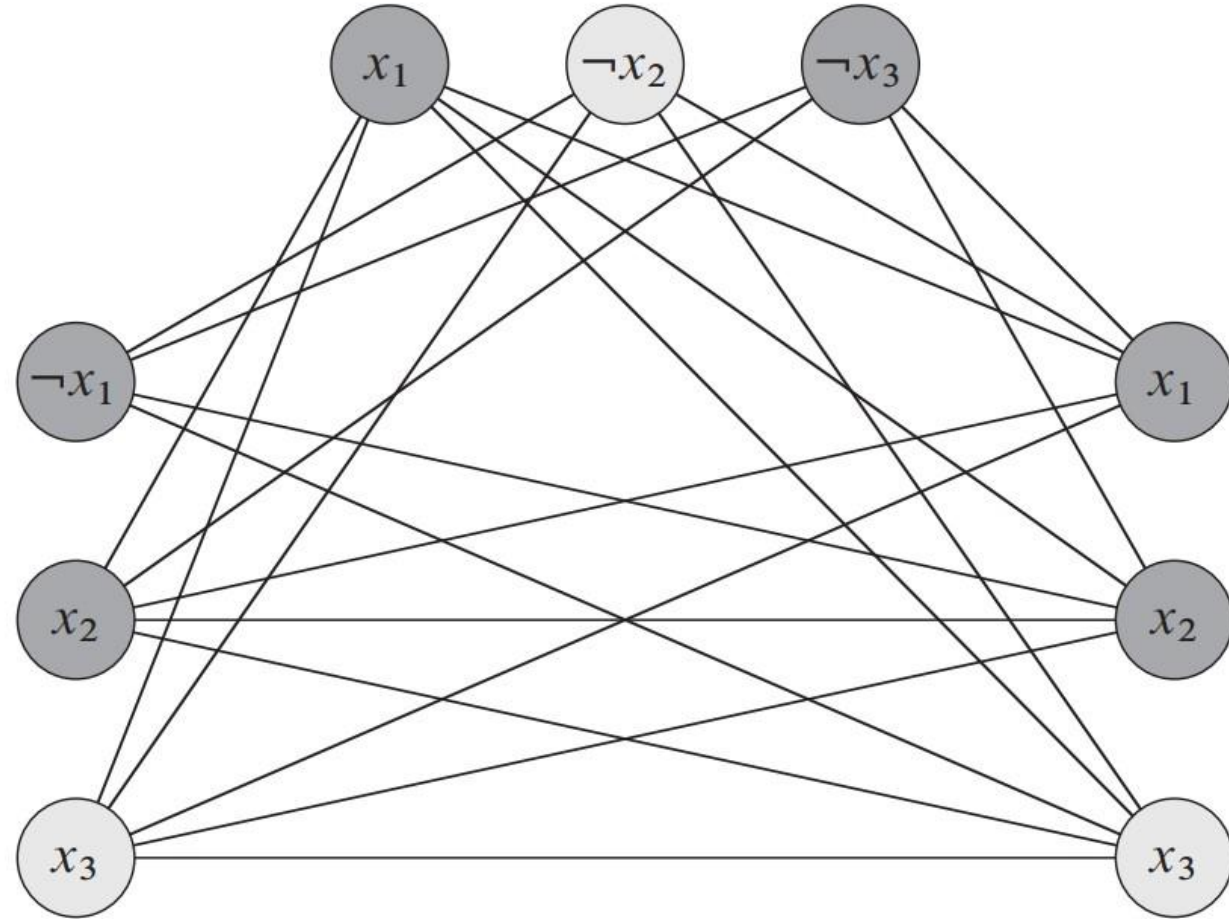
We construct the graph $G = (V, E)$ as follows. For each clause $C_r = (l_1^r \vee l_2^r \vee l_3^r)$ in ϕ , we place a triple of vertices v_1^r , v_2^r , and v_3^r into V . We put an edge between two vertices v_i^r and v_j^s if both of the following hold:

- v_i^r and v_j^s are in different triples, that is, $r \neq s$, and
- their corresponding literals are **consistent**, that is, l_i^r is not the negation of l_j^s .

We can easily build this graph from ϕ in polynomial time. As an example of this construction, if we have

$$\phi = (x_1 \vee \neg x_2 \vee \neg x_3) \wedge (\neg x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee x_3) ,$$

$$C_1 = x_1 \vee \neg x_2 \vee \neg x_3$$



$$C_2 = \neg x_1 \vee x_2 \vee x_3$$

$$C_3 = x_1 \vee x_2 \vee x_3$$

We must show that this transformation of ϕ into G is a reduction. First, suppose that ϕ has a satisfying assignment. Then each clause C_r contains at least one literal l_i^r that is assigned 1, and each such literal corresponds to a vertex v_i^r . Picking one such “true” literal from each clause yields a set V' of k vertices. We claim that V' is a clique. For any two vertices $v_i^r, v_j^s \in V'$, where $r \neq s$, both corresponding literals l_i^r and l_j^s map to 1 by the given satisfying assignment, and thus the literals cannot be complements. Thus, by the construction of G , the edge (v_i^r, v_j^s) belongs to E .

- For the above example

One possible satisfying assignment is

$x_1=1, x_2=1$ and $x_3=1$

$(1 \vee 0 \vee 0) \wedge (0 \vee 1 \vee 1) \wedge (1 \vee 1 \vee 1)$

Choosing

x_1 from c_1

x_2 from c_2

x_3 from c_3

Construct $V'=\{x_1, x_2, x_3\}$

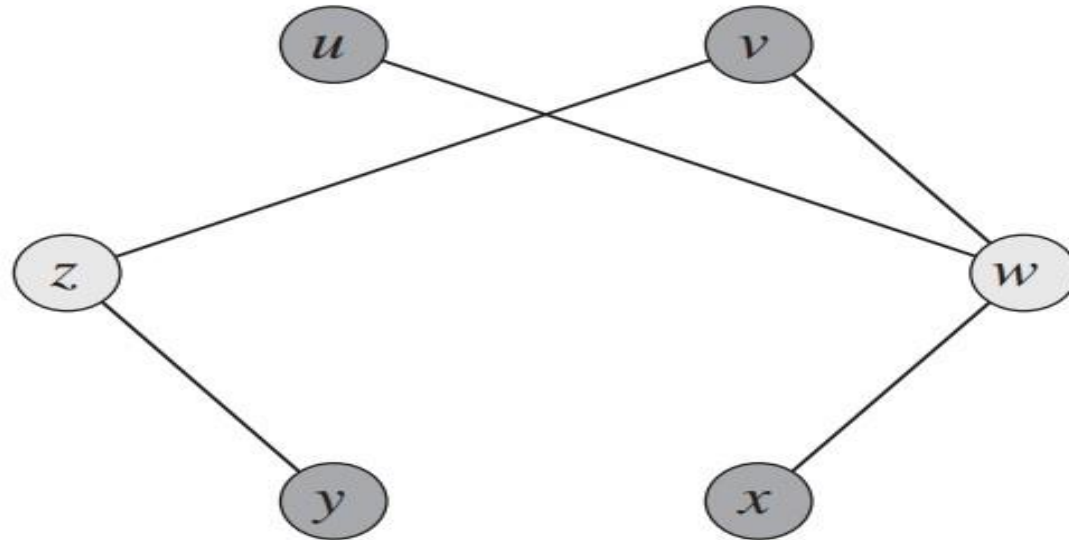
Check whether V' is a clique from the graph.

Problem

$$F = (x_2 \vee x_1 \vee x_3') \wedge (x_1' \vee x_2' \vee x_4) \wedge (x_2 \vee x_4' \vee x_3)$$

The vertex-cover problem

A *vertex cover* of an undirected graph $G = (V, E)$ is a subset $V' \subseteq V$ such that if $(u, v) \in E$, then $u \in V'$ or $v \in V'$ (or both). That is, each vertex “covers” its incident edges, and a vertex cover for G is a set of vertices that covers all the edges in E . The *size* of a vertex cover is the number of vertices in it. For example, the



The *vertex-cover problem* is to find a vertex cover of minimum size in a given graph. Restating this optimization problem as a decision problem, we wish to determine whether a graph has a vertex cover of a given size k . As a language, we define

$$\text{VERTEX-COVER} = \{ \langle G, k \rangle : \text{graph } G \text{ has a vertex cover of size } k \} .$$

The vertex-cover problem is NP-complete.

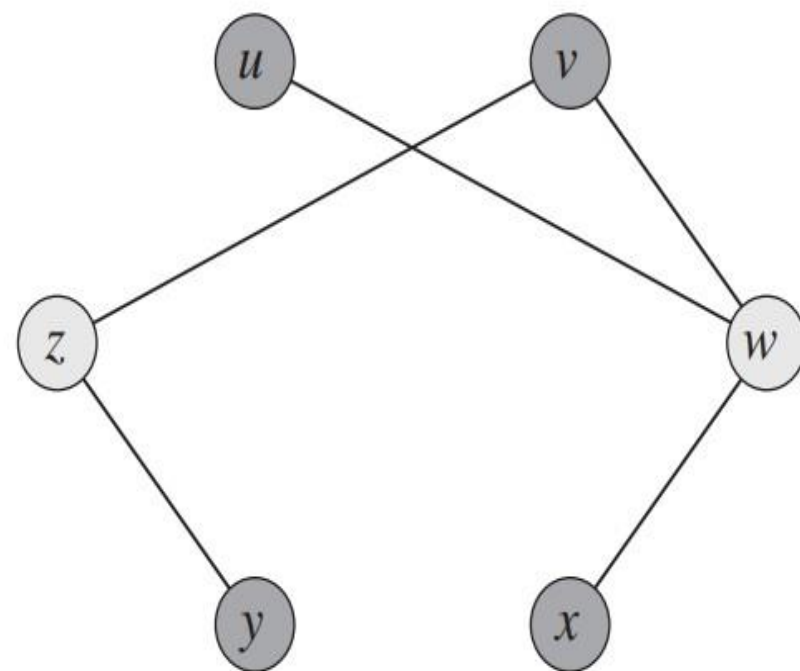
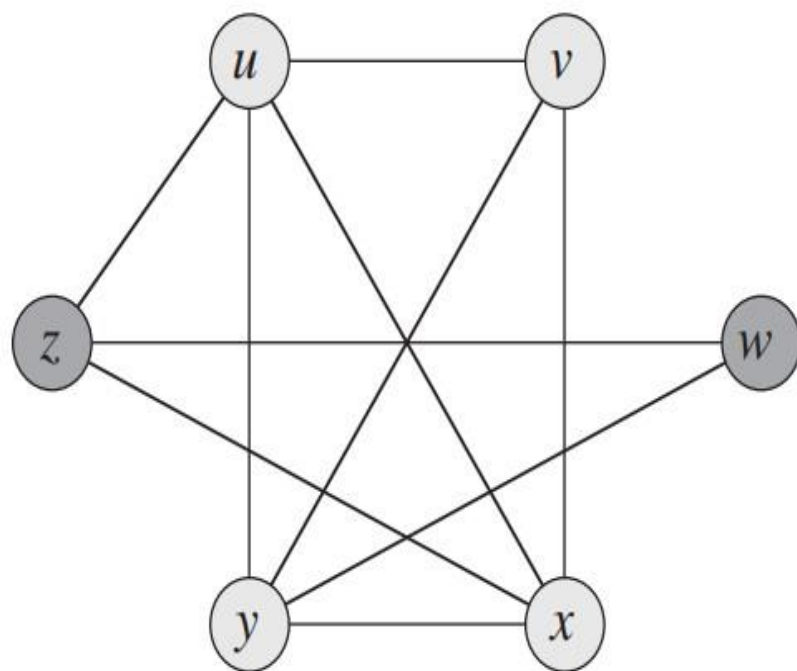
Proof:

- Show that **VERTEX-COVER $\in$ NP**. Suppose we are given a graph $G=(V, E)$ and an integer k , then it checks, for each edge $(u,v) \in E$, that $u \in V'$ or $v \in V'$.
- We can easily verify the certificate in polynomial time

Prove that the vertex-cover problem is **NP-hard** by showing that $\text{CLIQUE} \leq_P \text{VERTEX-COVER}$.

Step:1

as $\bar{G} = (V, \bar{E})$, where $\bar{E} = \{(u, v) : u, v \in V, u \neq v, \text{ and } (u, v) \notin E\}$. In other words, $\bar{G}$ is the graph containing exactly those edges that are not in G . Figure 34.15



Step 2:

G has a clique $V' \subseteq V$ with $|V'| = k$.
then $V - V'$ is the vertex cover of G'

Suppose that G has a clique $V' \subseteq V$ with $|V'| = k$. We claim that $V - V'$ is a vertex cover in $\overline{G}$. Let (u, v) be any edge in $\overline{E}$. Then, $(u, v) \notin E$, which implies that at least one of u or v does not belong to V' , since every pair of vertices in V' is connected by an edge of E . Equivalently, at least one of u or v is in $V - V'$, which means that edge (u, v) is covered by $V - V'$. Since (u, v) was chosen arbitrarily from $\overline{E}$, every edge of $\overline{E}$ is covered by a vertex in $V - V'$. Hence, the set $V - V'$, which has size $|V| - k$, forms a vertex cover for $\overline{G}$.

Travelling Salesman Problem

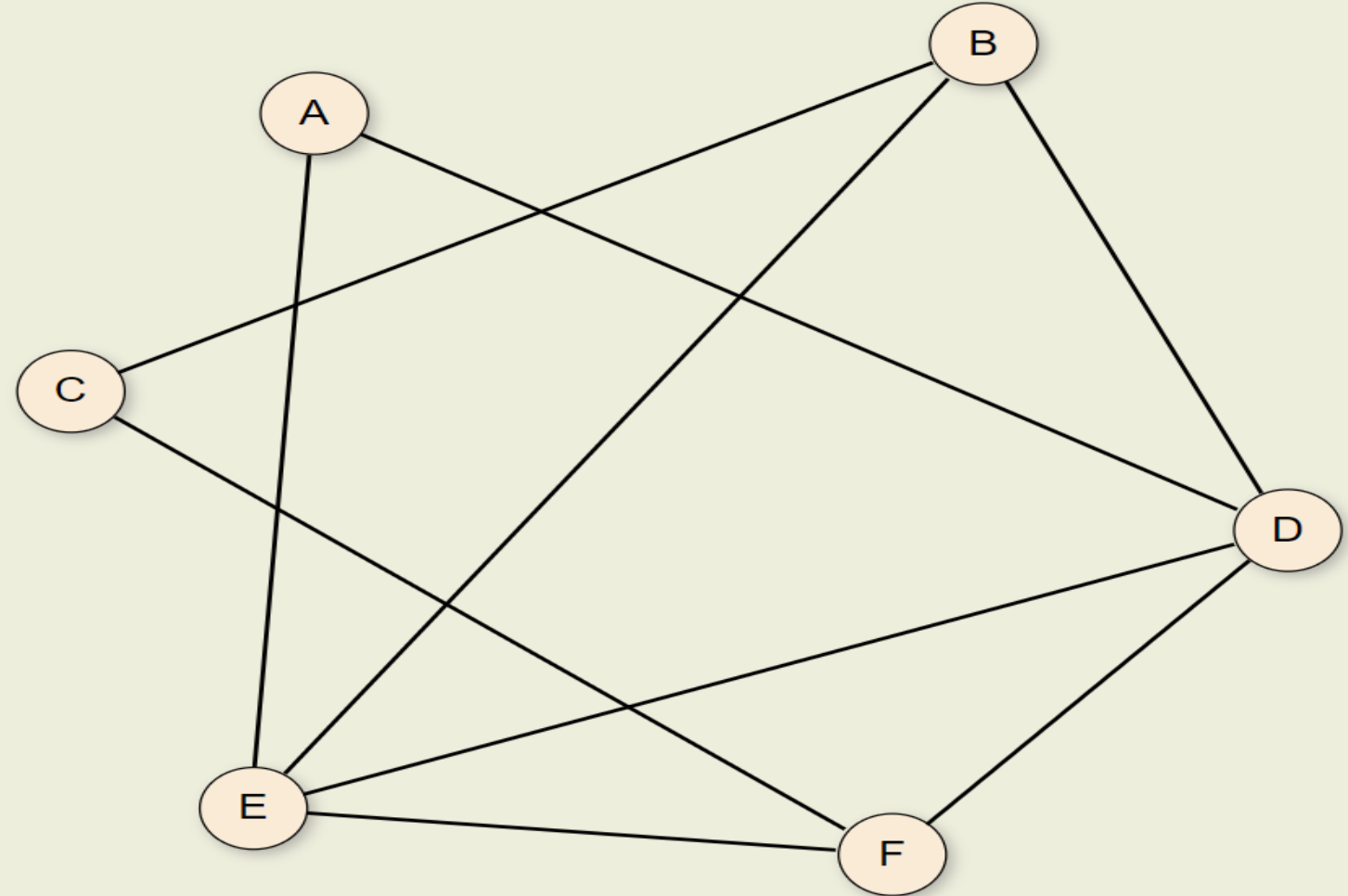
Hamiltonian Cycle and Traveling Salesman problems

For a given graph $G = (V, E)$, the Hamiltonian Cycle problem is to find whether G contains a Hamiltonian Cycle that is, a cycle that passes through all the vertices of the graph exactly once.

For a given weighted graph $G' = (V', E')$, with non-negative weights, and integer k' , the Traveling Salesman problem is to find whether G' contains a simple cycle of length $\leq k'$ that passes through all the vertices. [Length of a cycle is the sum of weights of all the edges in the cycle].

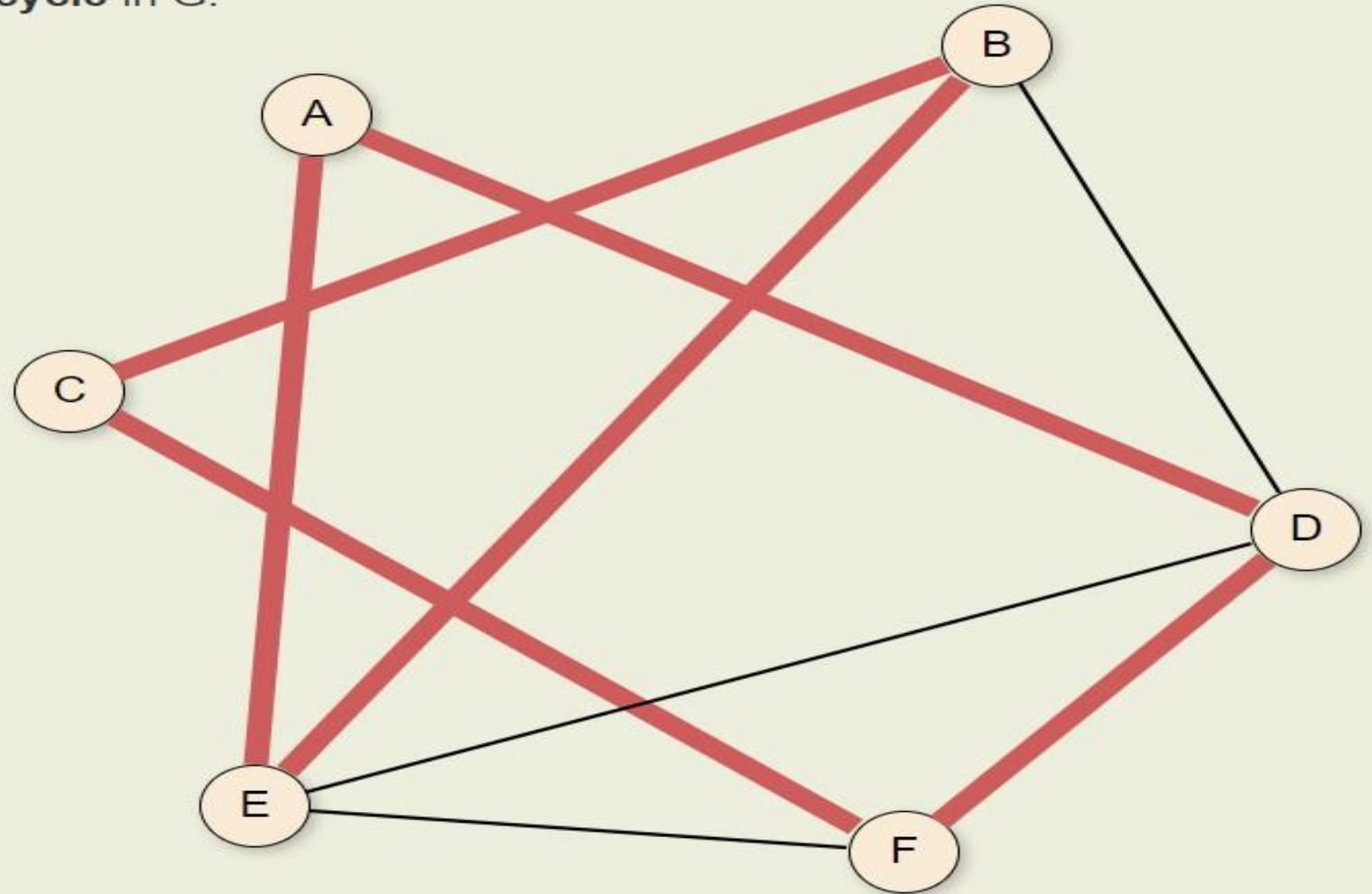
Example graph

Let this graph G be an input to the Hamiltonian Cycle problem



Example:

G' has a cycle passing through all vertices exactly once with length ≤ 0
This cycle is a **Hamiltonian cycle** in G .



The traveling-salesman problem is NP-complete

show that **TSP belongs to NP**

- Given an instance of the problem, we use as a certificate the sequence of n vertices in the tour.
- The verification algorithm checks that this sequence contains each vertex exactly once, sums up the edge costs, and checks whether the sum is at most k .
- This process can certainly be done in polynomial time

prove that TSP is **NP-hard**

show that HAM-CYCLE $\leq_P$ TSP

Reduction of Hamiltonian Cycle Problem to Traveling Salesman Problem

To reduce the Hamiltonian Cycle Problem to the Traveling Salesman problem for a given graph $G = (V, E)$, complete the graph G , by adding edges between all pairs of vertices that were not connected in G

Let the new graph be $G' = (V', E')$ where $V' = V$ and $E' = \{(u, v)\}$ for any $u, v \in V'$.

For edges in G' that were also present in G , we assign a weight 0.

For other edges we assign weight 1

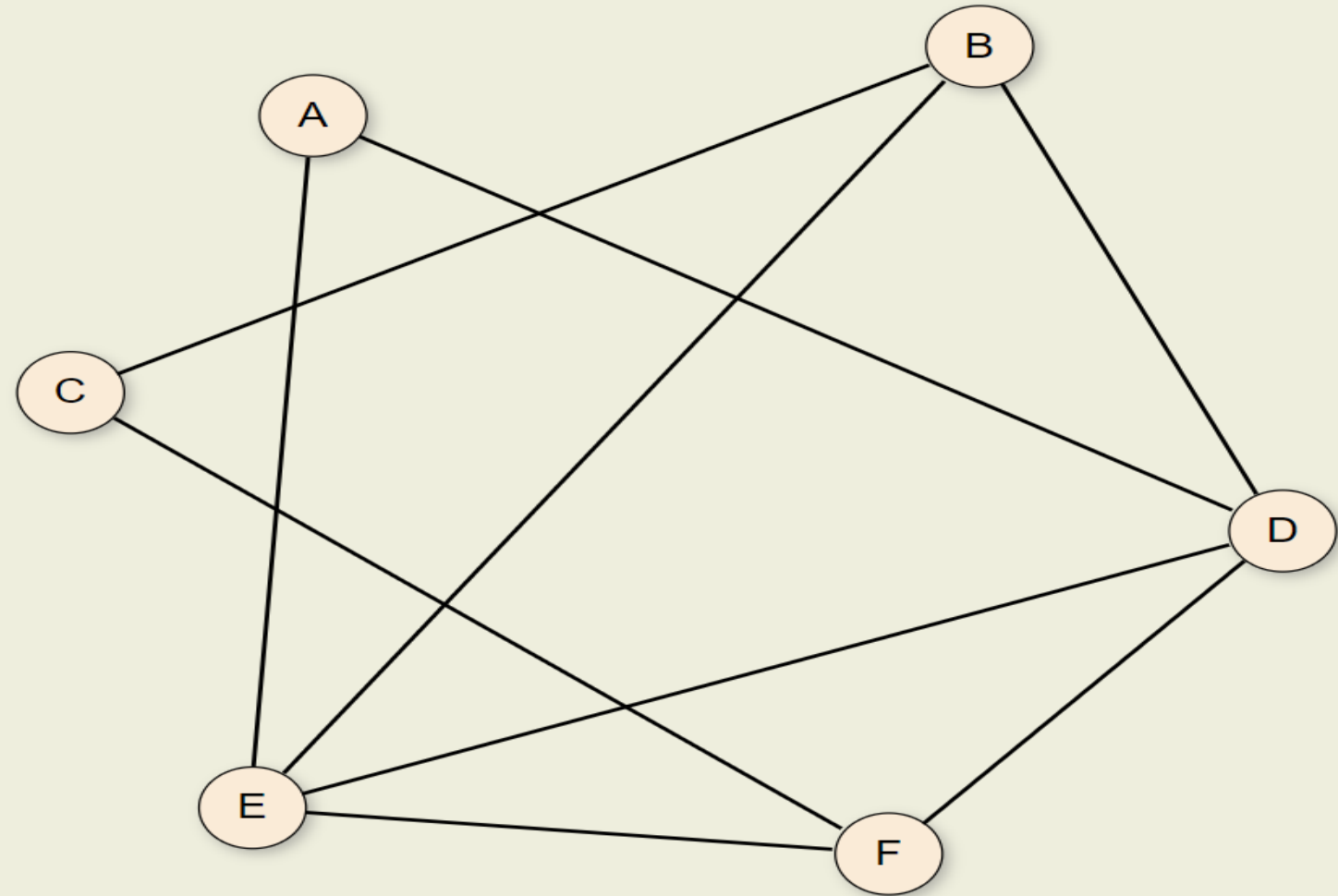
that is, $\forall e = (u, v) \in E'$,

$$W(e) = 0, \text{ if } (u, v) \in E$$

$$W(e) = 1, \text{ if } (u, v) \notin E$$

Example graph

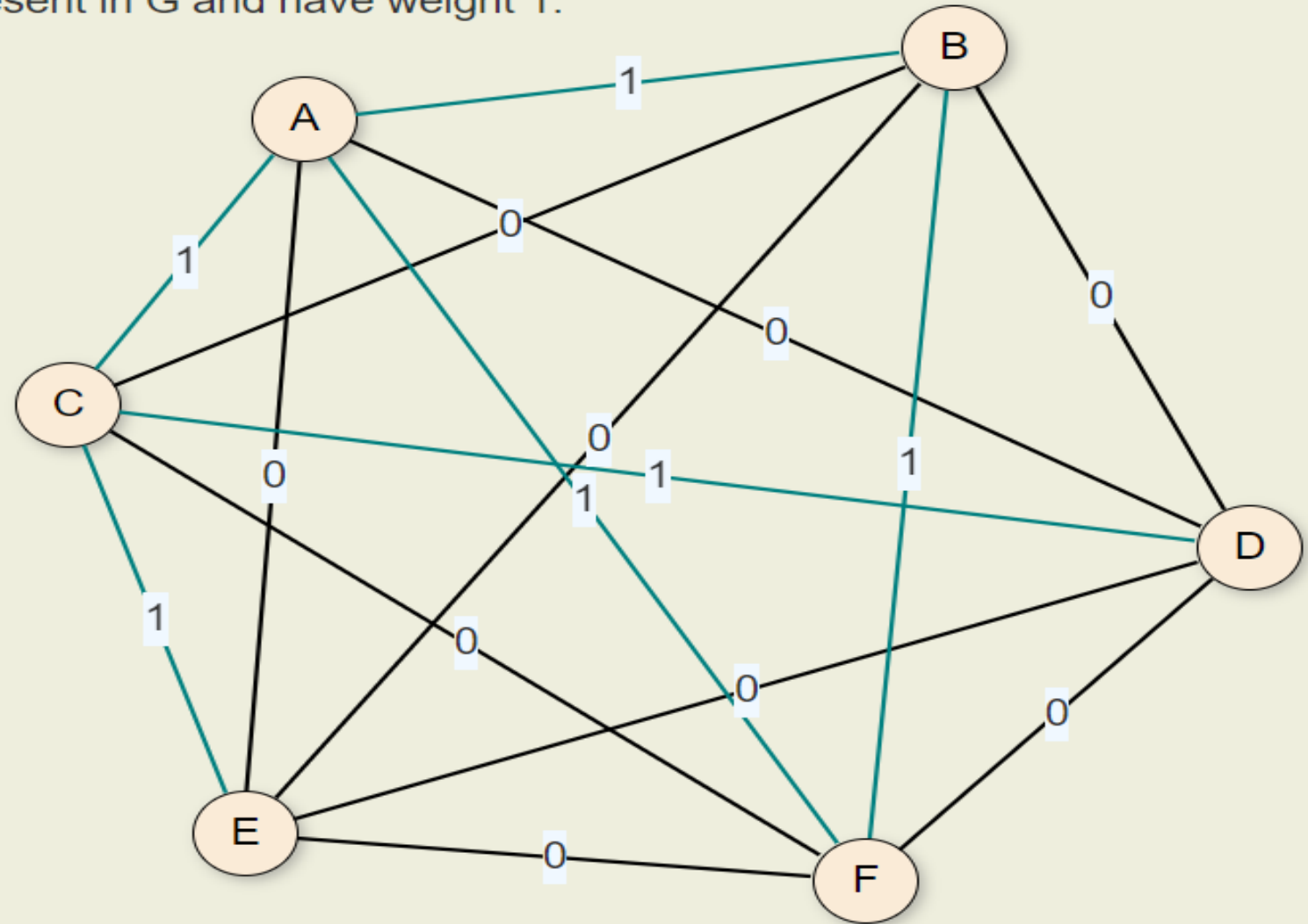
Let this graph G be an input to the Hamiltonian Cycle problem



Example graph

The constructed graph G' is as below.

The blue edges were not present in G and have weight 1.



Hamiltonian Cycle problem reduced to an instance of Traveling Salesman Problem

The graph G has a Hamiltonian Cycle if and only if there exists a cycle in G' passing through all vertices exactly once, and that has a length ≤ 0 (i.e. has a solution for the instance of Traveling Salesman Problem where $k = 0$)

Hamiltonian Cycle problem reduced to an instance of Traveling Salesman Problem

The graph G has a Hamiltonian Cycle if and only if there exists a cycle in G' passing through all vertices exactly once, and that has a length ≤ 0 (i.e. has a solution for the instance of Traveling Salesman Problem where $k = 0$)

1. If there is a cycle that passes through all vertex exactly once, and has length ≤ 0 in graph G' , the cycle contains only edges that were originally present in graph G . (The new edges in G' have weight 1 and hence can not be part of a cycle of length ≤ 0 .)

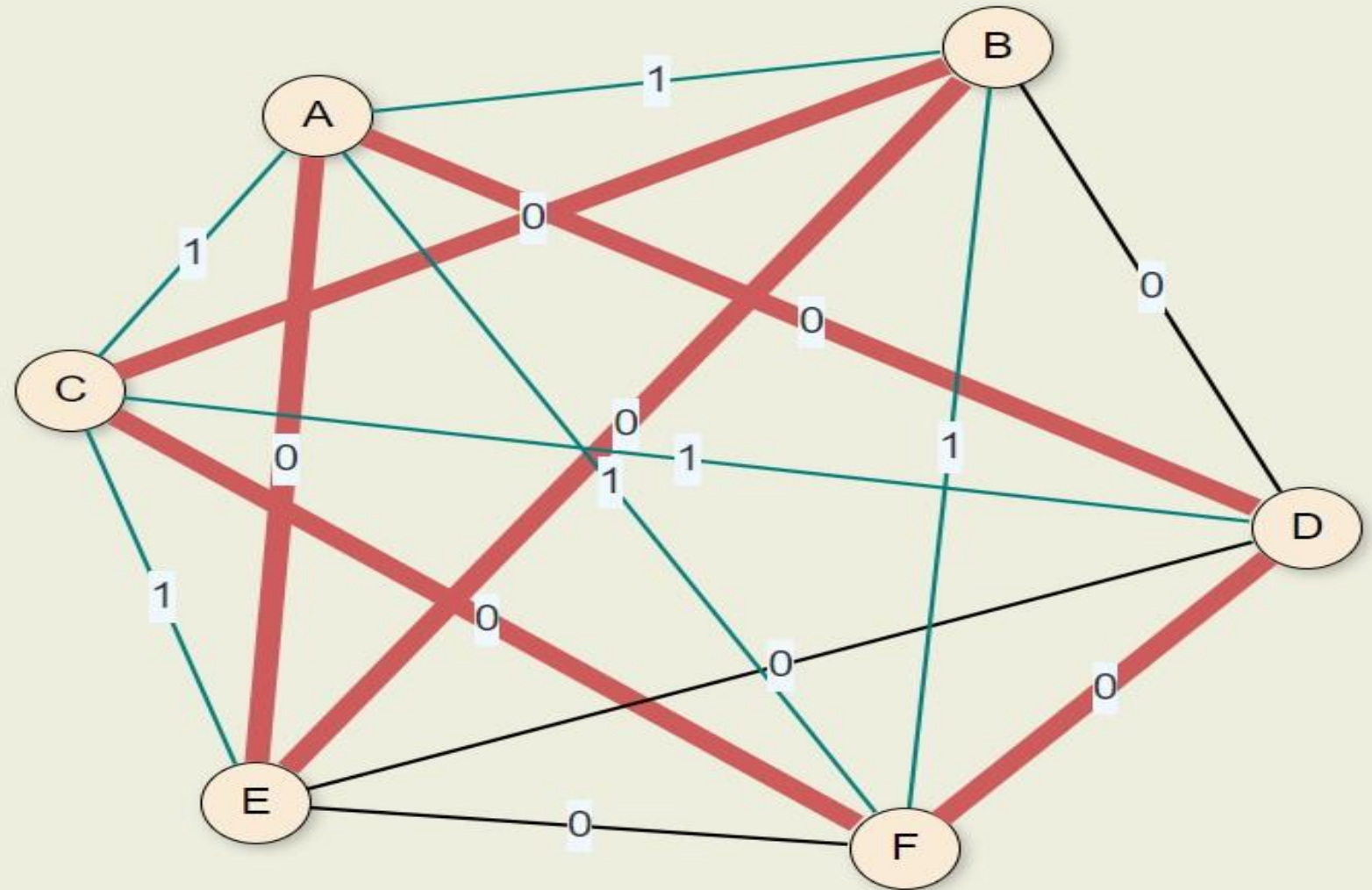
Hence there exist a Hamiltonian cycle in G

2. If there exists a Hamiltonian Cycle in the graph G , it forms a cycle in G' with length $= 0$, since a weights of all the edges is 0.

Hence there exists a solution for Traveling Salesman Problem in G' with length ≤ 0

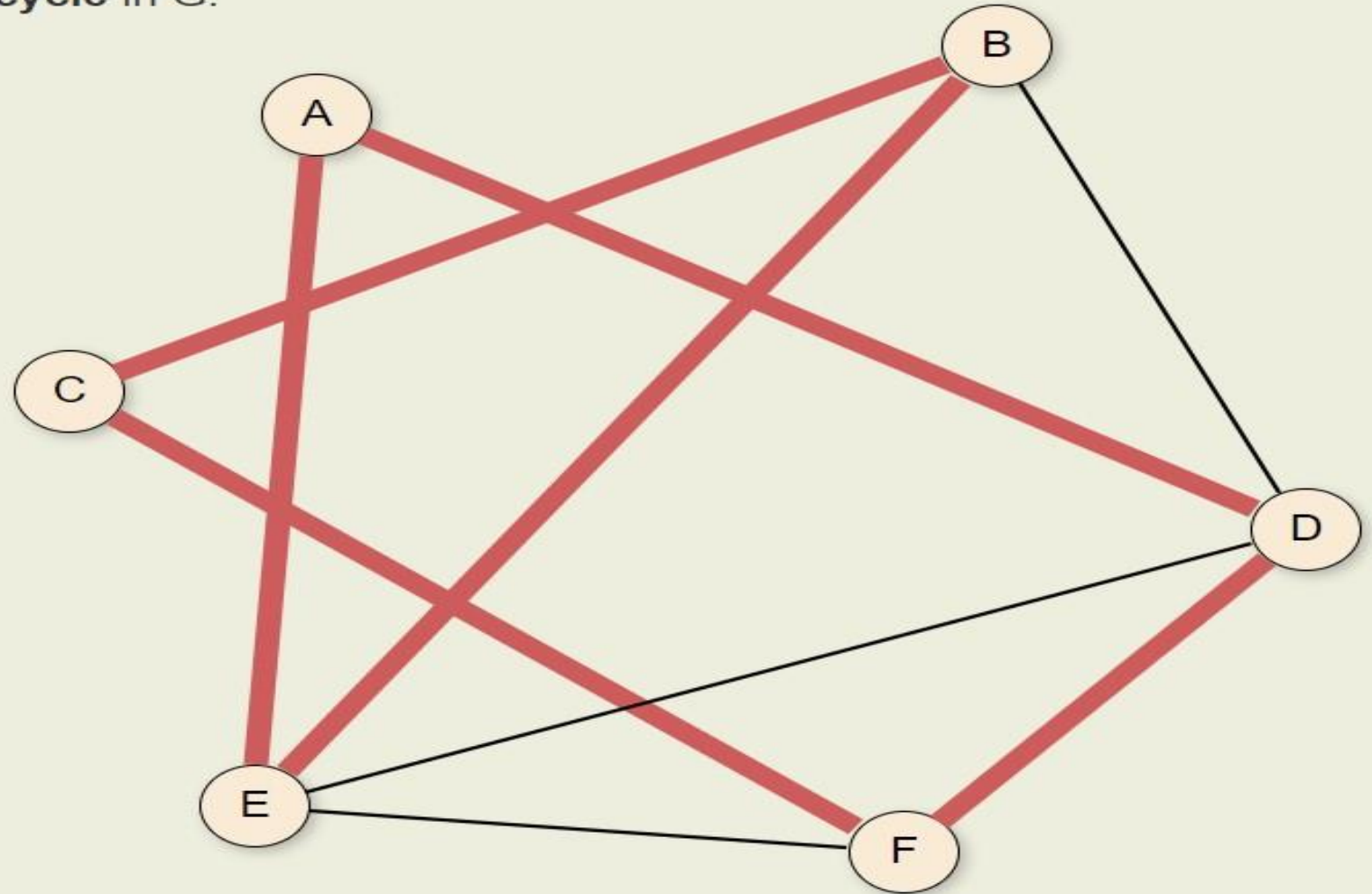
Example:

G' has a cycle passing through all vertices exactly once with length ≤ 0



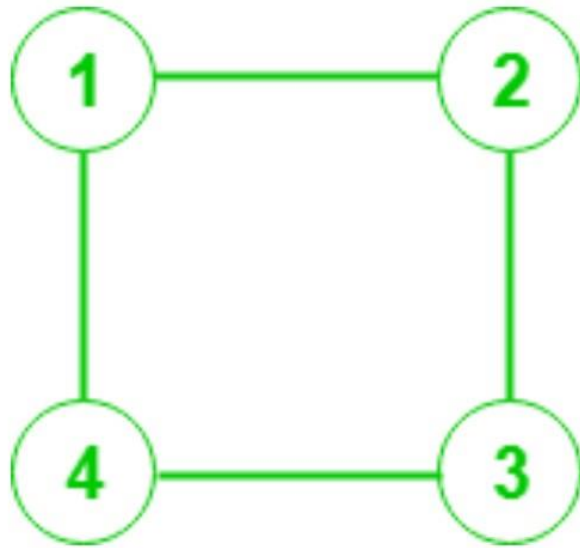
Example:

G' has a cycle passing through all vertices exactly once with length ≤ 0
This cycle is a **Hamiltonian cycle** in G .



Independent Set

An Independent Set S of graph $G = (V, E)$ is a set of vertices such that no two vertices in S are adjacent to each other. It consists of non-adjacent vertices.



$$I_1 = \{1, 3\} \text{ and } I_2 = \{2, 4\}$$

INDEPENDENT SET (IS) is NP-complete

- First, **IS is in NP**, since given any set S we can check in polytime that S is independent and that $|S| = k$.
- So for a yes instance, we simply use an independent set of size k .
- And for a no instance, it is clear that no such set exists.
- Therefore IS is in NP.

flag=true

For every pair $\{u, v\}$ in the subset V' :

 Check that these two don't
 have an edge between them

 If there is an edge,
 set flag to false and break

If flag is true:

 Solution is correct

Else:

 Solution is incorrect

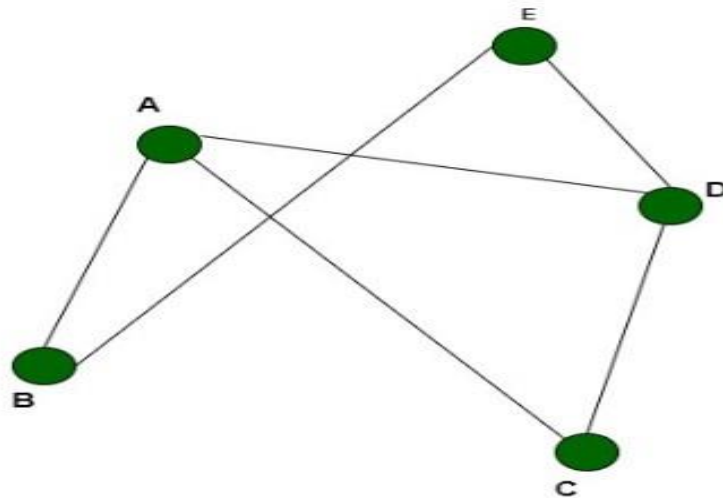
show that IS is **NP-Hard**

- In order to prove that the Independent Set problem is NP-Hard, we will perform a reduction from a known NP-Hard problem to this problem
- **Clique $\leq$ Independent Set**

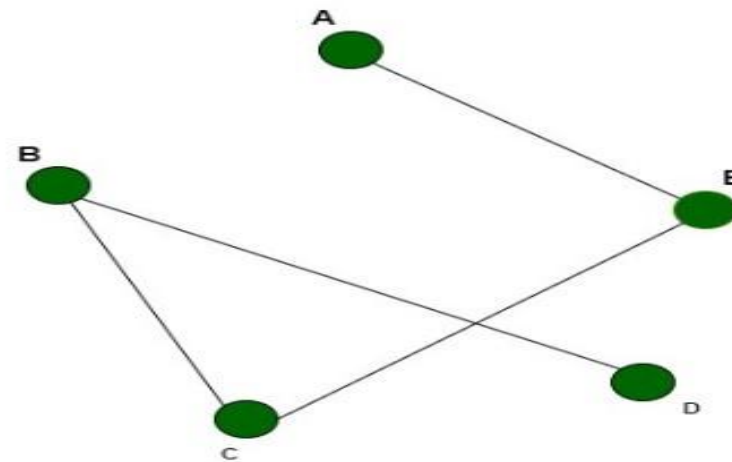
Every instance of the clique problem consisting of the graph G (V, E) and an integer k can be converted to the required graph G' (V', E') and k' of the Independent Set problem.

Step 1:

Construct graph G' is the complementary graph of G



$G \cong$
Clique of size 3 {A C D}



$G' \cong$
Independent Set of size 3 {A C D}

- Let us assume that the graph G contains a clique of size k .
- The presence of clique implies that there are k vertices in G , where each of the vertices is connected by an edge with the remaining vertices.
- This further shows that since these edges are contained in G , therefore they can't be present in G' .
- As a result, these k vertices are not adjacent to each other in G' and hence form an Independent Set of size k .